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NAME: R.BalaMurugan STD.: IT SEC.: A ROLL NO.: 026 SUB.: Software construction

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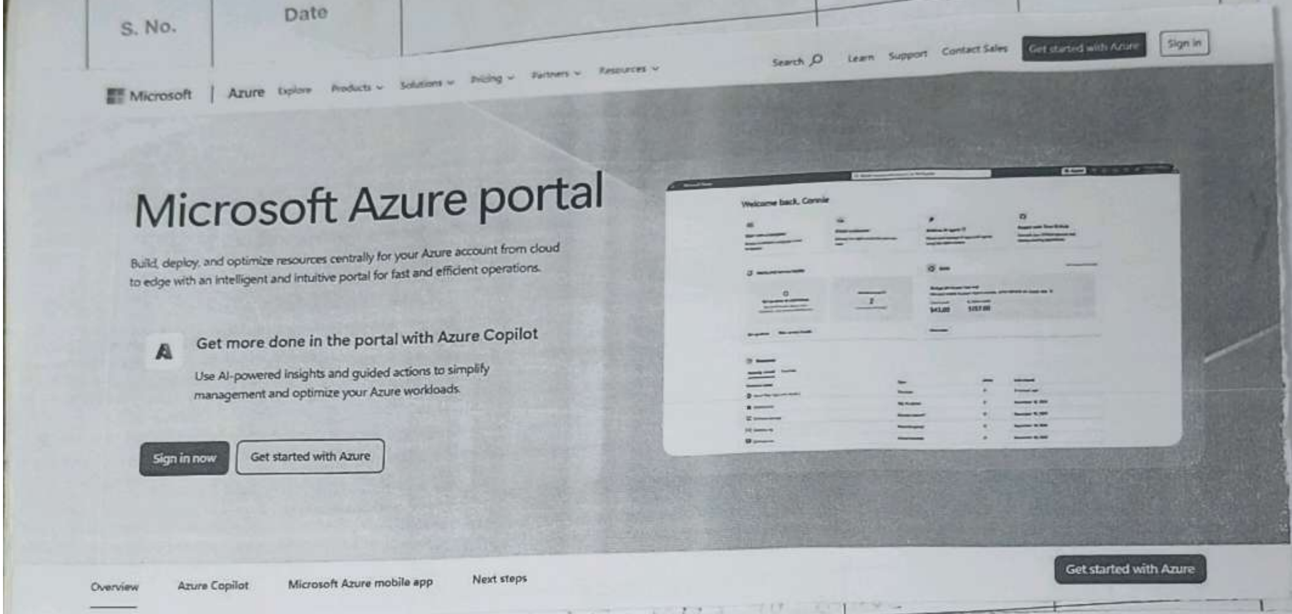
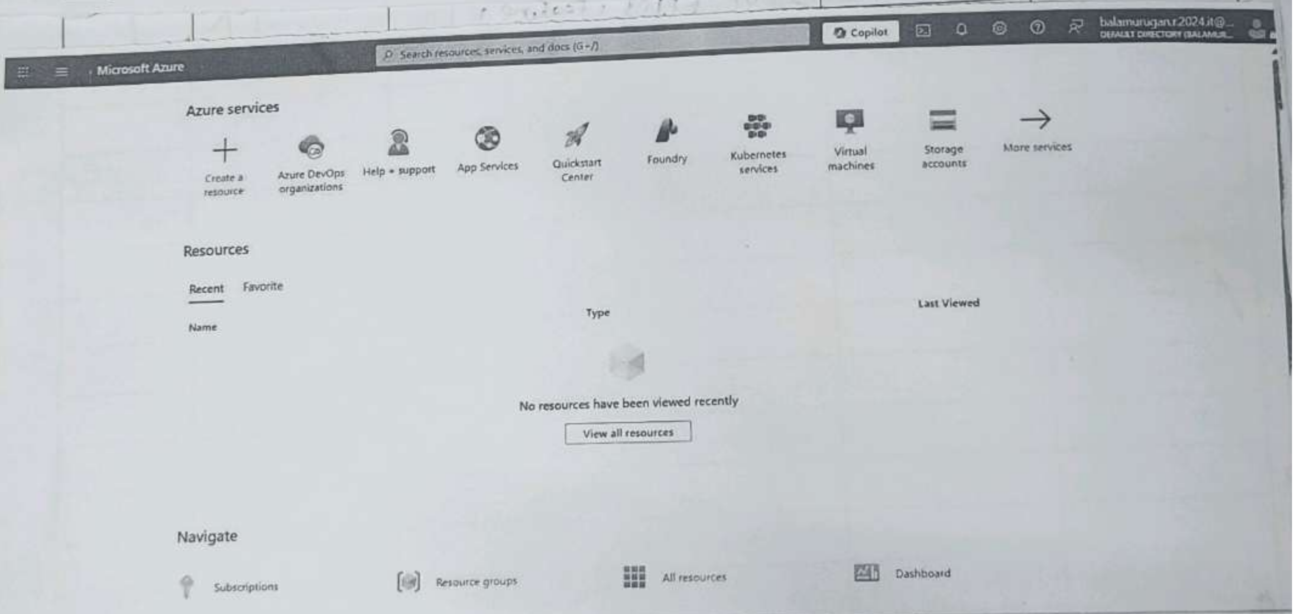




EXP NO : 1 AZURE DEVOPS ENVIRONMENT SETUP

AIM:

To setup and access the Azure DevOPS environment by creating an organization through the Azure Portal

INSTALLATION :

1) Open your web browser and go to the Azure website:

https://azure.microsoft.com/en-us/get-started/azure-portal

* sign in using your Microsoft account credentials
* If you don't have a microsoft account, create one using the signup page

Observation :

-> The microsoft Azure Portal homepage is displayed
-> The sign in option is visible on the screen
-> options to explore & get started with Azure services are available

2) Azure Home Page :

After signing in, the Azure home dashboard is displayed with various services and tools

observation :

-> The Azure dashboard is displayed after successful again
-> A search bar is available at the top to access service
-> Different Azure services & recent resources are visible.

3) Open DevOPS Environment :

Open DevOPS environment in the Azure Platform by typing Azure DevOPS organizations in the search bar

Observation :

-> The search bar is used to enter Azure Devops organiz
-> A list of related services appears in the dropdown
-> Azure Devops option is visible for selection

4) Create organization:

click on the My Azure DevOps organization link to create an organization. you will be taken to the Azure DevOps organization Home page.

Observation :-

-> The Azure DevOps page is displayed

-> options to create or access an organization are available

-> A new organization is created successfully

-> The organization dashboard/ home page is opened

Result :

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal

EXP NO : 2 : AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

AIM :

To setup an Azure Devops Project for efficient collaboration & agile work management

PROCEDURE :

1) create an Azure account

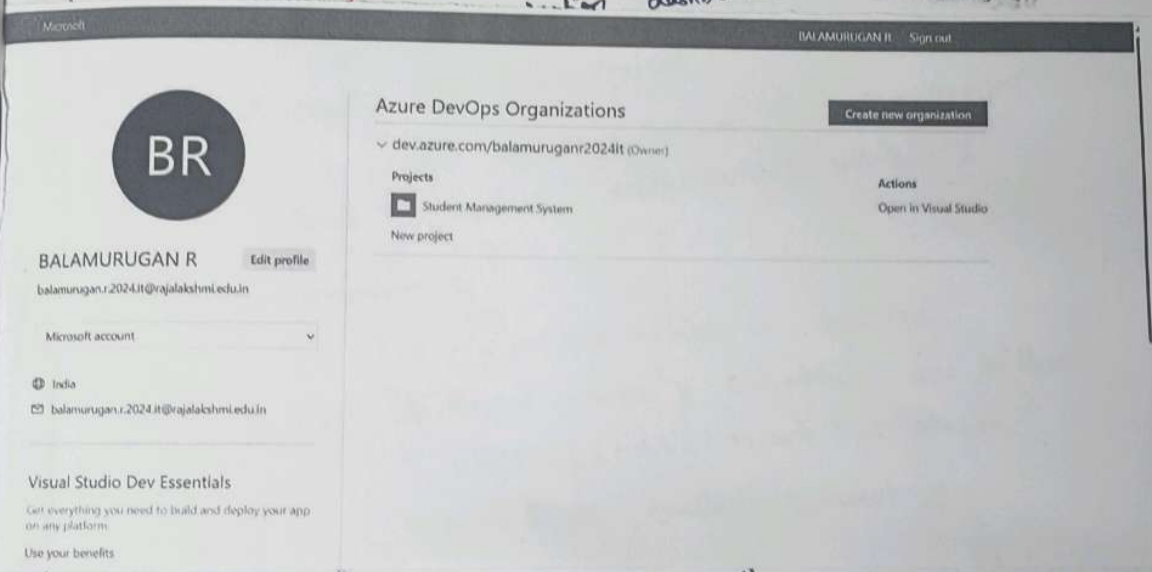
Login to Azure Devops using your microsoft account

Observation :

-> Azure Devops portal is opened successfully.

-> User logged in using valid credentials

-> ...tion dashboard is displayed



2) Switch organization (if needed)

Ensure you are in the correct organization by selecting it from the top menu

Observation :

-> organization name is visible at the top

-> User can switch between multiple organizat

-> correct organization dashboard is displayed

4) View Project Dashboard

Open the created project dashboard

Observation :

-> project dashboard is displayed

-> Options like Boards, Repos, Pipelines are available

-> project overview & details are visible.

5) Manage user stories

Go to Boards -> work items -> Add work item -> user story & enter details

Observation:

-> Boards section is opened successfully

-> work item option is available

-> user story form is displayed

-> user story details are entered & saved

Result :

Successfully created an Azure DevOps Project and managed user stories using agile workflow.

EXP NO 3: SETTING UP EPICS, FEATURES AND USER STORIES FOR PROJECT PLANNING"

AIM:

To learn about how to create epics, user stories, features, and backlogs for the assigned project

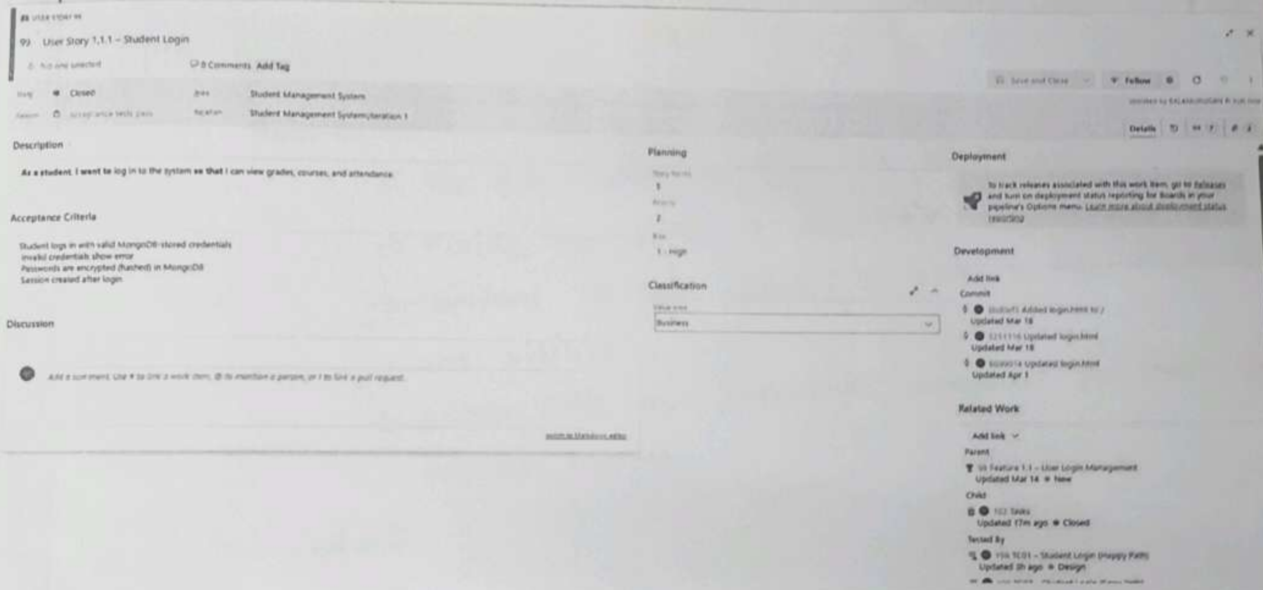
create Epic, Features, user stories, Task

Observation :

→ The Azure Devops Boards section is displayed

→ The backlog contains different work items

→ Work items are shown with details like ID, state & assigned user.



Observation :

→ The Feature work item page is displayed

→ Fields such as title, description & priority are available

→ features are created under Epics.

→ Features help in organizing project functionality

8) Fill in user story details

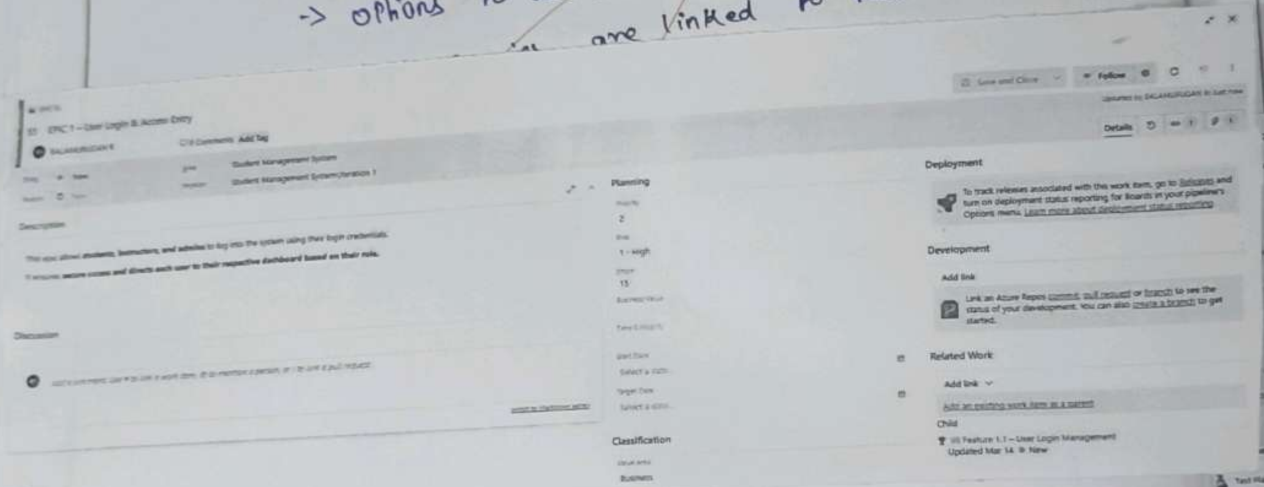
Observation :

-> The user story form is displayed

-> fields for description & acceptance criteria are available

-> Options to assign users & update status are visible

are linked to features & used



Result :

Thus the creation of EPICs, Features, user stories and Tasks has been completed successfully.

EXP NO : 4 SPRINT PLANNING

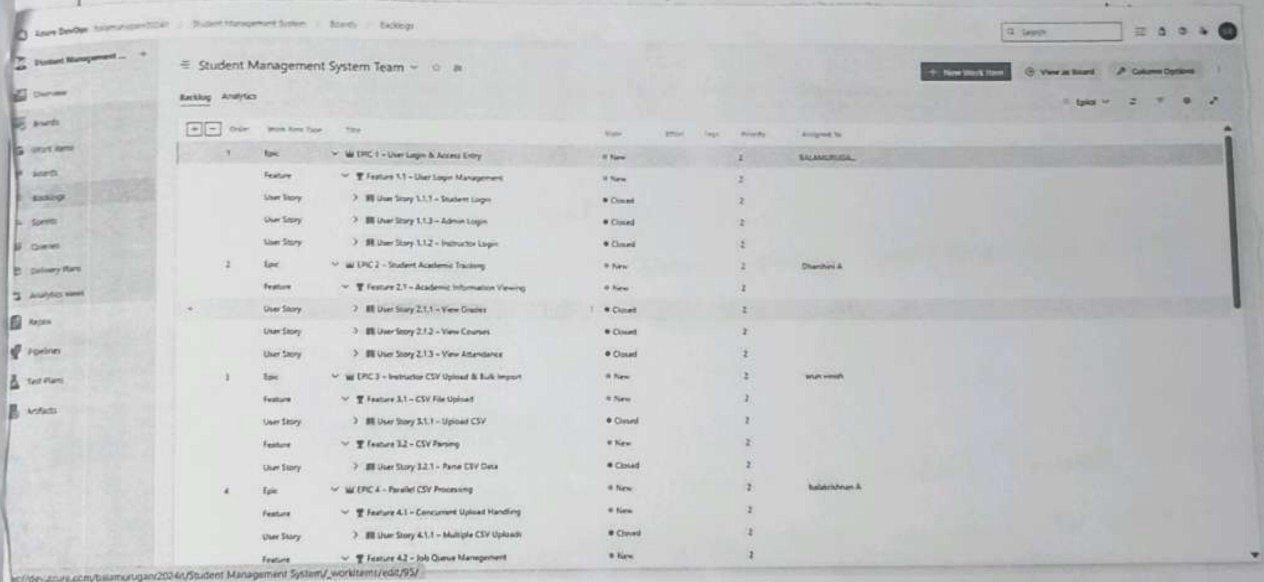
AIM :

TO assign user stories to specific sprints for the student data management system project.

Sprint Planning

Sprint 1 :

Observation :



Azure DevOps / Student Management System / Boards / Backlogs

Student Management System Team

Backlog Analytics

Order	Work Item Type	Title	State	Effort	Tags	Priority	Assigned To
1	Epic	EPIC 1 - User Login & Access Entry	New			2	BALAMURUGA...
	Feature	Feature 1.1 - User Login Management	New			2	
	User Story	User Story 1.1.1 - Student Login	Closed			2	
	User Story	User Story 1.1.3 - Admin Login	Closed			2	
	User Story	User Story 1.1.2 - Instructor Login	Closed			2	
2	Epic	EPIC 2 - Student Academic Tracking	New			2	Dharshini A
	Feature	Feature 2.1 - Academic Information Viewing	New			2	
	User Story	User Story 2.1.1 - View Grades	Closed			2	
	User Story	User Story 2.1.2 - View Courses	Closed			2	
	User Story	User Story 2.1.3 - View Attendance	Closed			2	
3	Epic	EPIC 3 - Instructor CSV Upload & Bulk Import	New			2	[illegible]
	Feature	Feature 3.1 - CSV File Upload	New			2	
	User Story	User Story 3.1.1 - Upload CSV	Closed			2	
	Feature	Feature 3.2 - CSV Parsing	New			2	
	User Story	User Story 3.2.1 - Parse CSV Data	Closed			2	
4	Epic	EPIC 4 - Parallel CSV Processing	New			2	balakrishnan A
	Feature	Feature 4.1 - Concurrent Upload Handling	New			2	
	User Story	User Story 4.1.1 - Multiple CSV Uploads	Closed			2	
	Feature	Feature 4.2 - Job Queue Management	New			2	

://dev.azure.com/balamurugan2024/Student Management System/_workitems/edit/95/

-> Tasks are organized based on [illegible] priority

-> The sprint board is used to track progress of student management features

Sprint 3 :

Observation :

-> user stories such as view student details & search student information are included

-> work items are grouped under data retrieval operation

-> The sprint board reflects tasks progress & current status.

Sprint 4 :

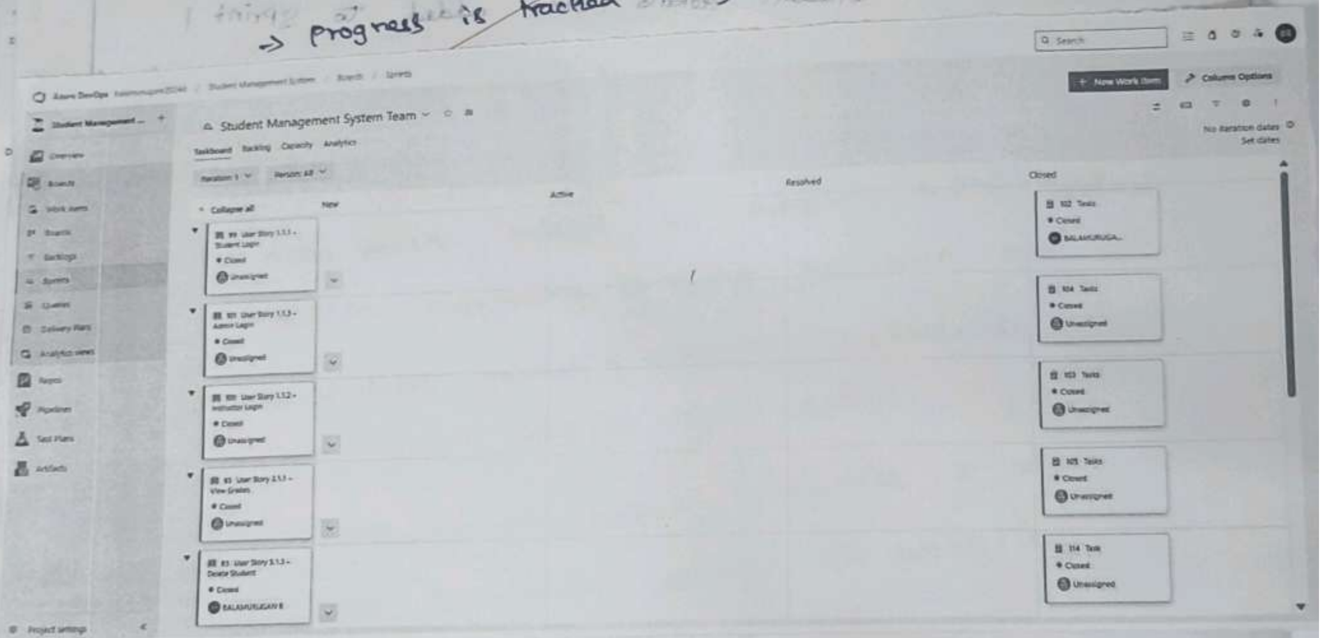
Observation :

-> user stories from Search & filter and data validation & Error Handling are included.

-> Focus is on retrieving student information and validating input data

-> Tasks are grouped under data Processing & validation operation

-> Progress is tracked using sprint board workflow.



until completion.

Result :

Sprints are Successfully planned & user Stories are assigned effectively for the Student management system project.

EXPNO 5: POKER ESTIMATION

AIM:

To estimate effort for user stories using the poker Estimation Technique in the student management system

Poker Estimation:

Observation:

-> user stories ~~selected~~ from different epics are listed in Azure DevOps backlog

-> Each story is analyzed by team member

-> Story points are assigned based on complexity and development effort.

1) Selecting user stories:

Observation:

User stories selected from:

-> User login & access Entry

-> Student data management

-> Search & filter

-> Reporting & Dashboard

2) Estimation using poker cards

Observation:

-> Team members use poker cards to estimate effort

-> Each member gives estimation independently

-> Estimates vary based on ~~everything~~ understanding of complexity.

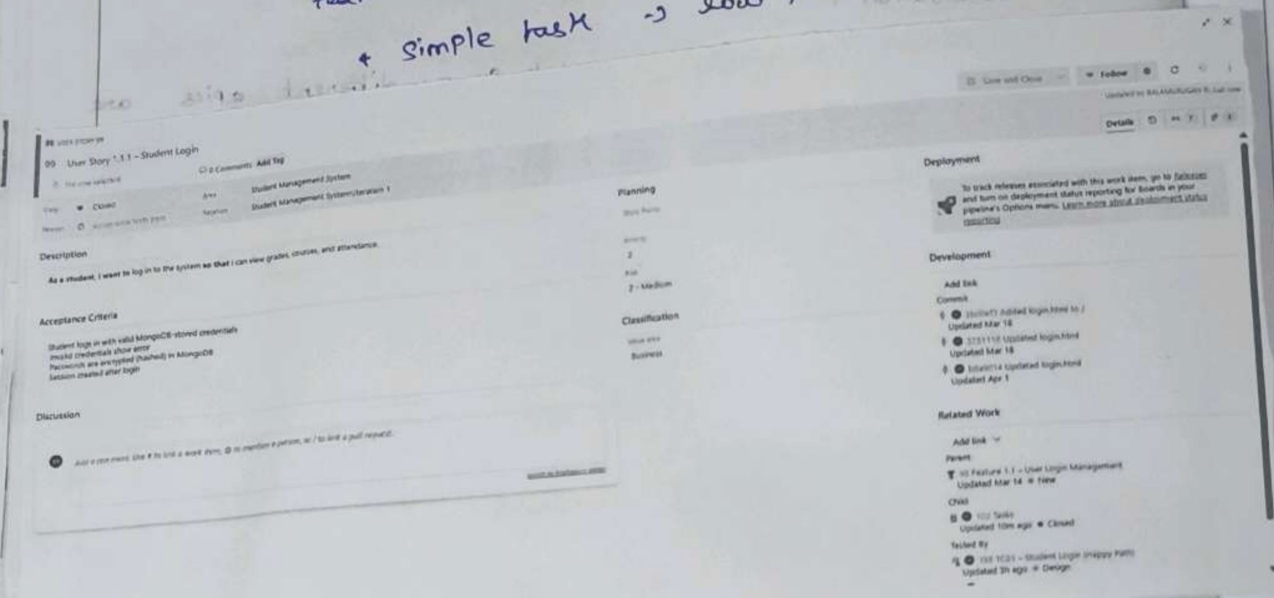
3) Finalizing story point

Observation :

→ All estimates are compared & discussed.

→ Final ~~year~~ story points are decided based on team agreement.

+ Simple task → low points (1 - 3)



4	Parallel processing	8
5	CRUD Management	5
6	Course & Grade	5
7	Validation	2
8	Search & filter	3
9	Admin monitoring	5
10	Reporting	8

5) Updating in Azure DevOps

observation :

→ Final story points are updated in the Effort field of each work item

→ Backlog reflects estimated values

→ This helps in sprint planning & workload distribution.

Result :

User stories are successfully estimated using Poker Estimation technique for the Student Management System project, improving planning accuracy & team collaboration.

EXP NO: 6

Designing class and Sequence Diagram for Project Architecture

AIM:

To design a class diagram & sequence diagram for the student data management system.

class Diagram:

A class Diagram represent the static structure of the system. It shows

* classes
* Attributes
* Method
* Relationships

classes used:

-> Student -> course
-> Grade -> Admin

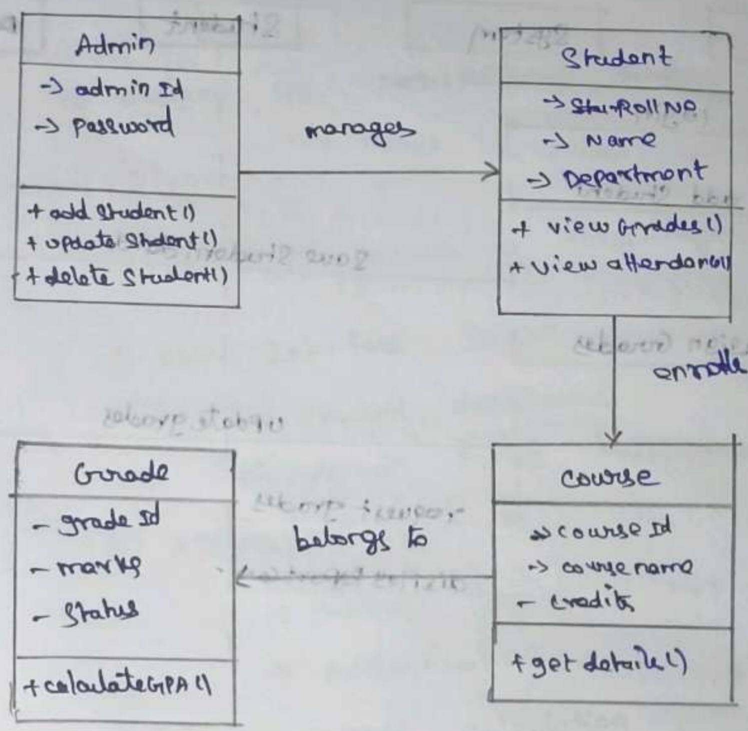
Description:

-> Student: stores student details like ID, name, department, contact

-> course: contains details such as ID, name, credits.

-> Grades: stores marks & result status

-> Admin: manages student and course records class Diagram :



Sequence Diagram :

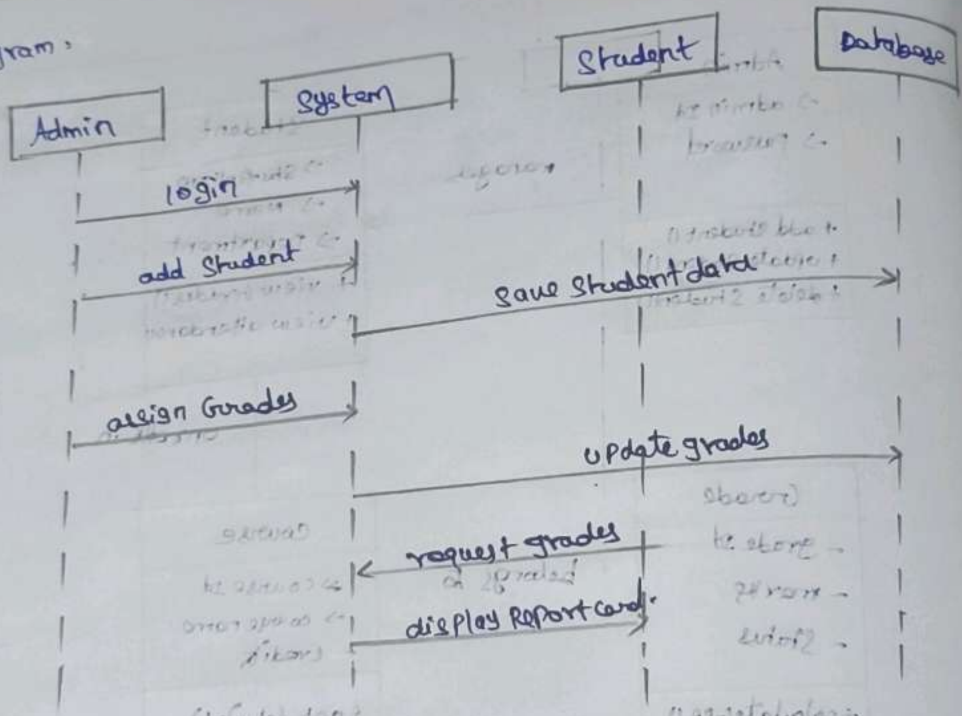
Description :

This diagram shows the flows of operation in the system.

Steps :

1) Admin logs into System
2) Admin adds a student
3) System saves data to databases
4) Admin assigns grades
5) System updates grades
6) Student request grades
7) System display report card.

Diagram:



Result:

The class & Sequence diagram are diagonal successfully for the ~~do~~ Student data management System

EXP NO 7: Designing Architectural & ER Diagram

AIM

To design the Architectural diagram & ER diagram

Architectural Diagram :

The system follows a three-layer Architectural

1) User Interface layer

→ Student dashboard

→ Admin panel

2) Application Layer

→ Business layer

→ Authentication

→ Grade calculation

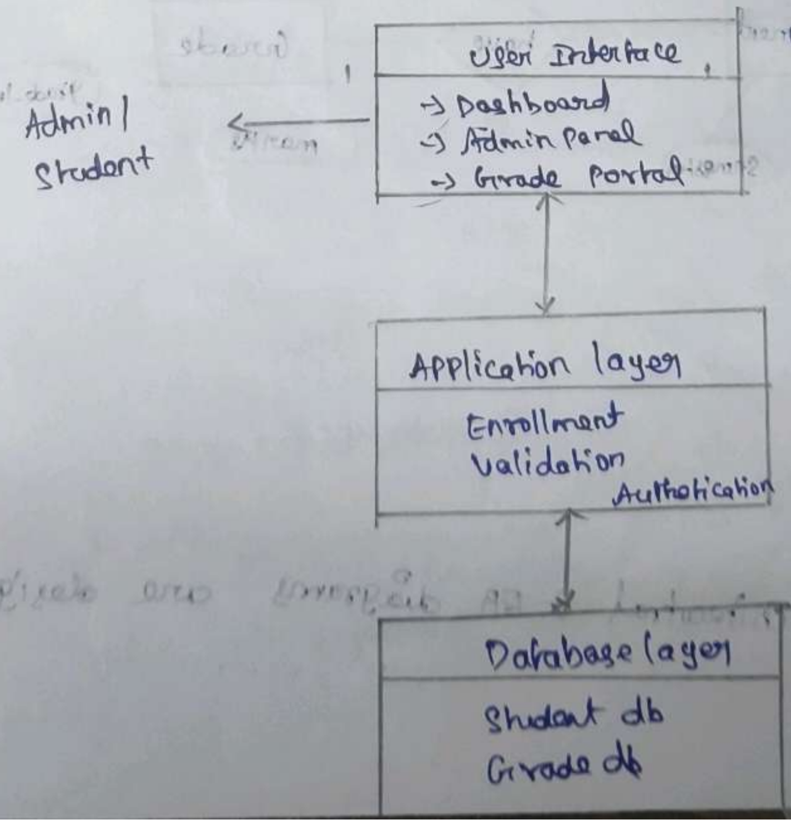
3) Database layer

→ Student database

→ Course database

→ Grade database

Flow diagram :



ER Diagram

Entities :

-> Student (Student-id, name, dob, department)

-> Course (course_id, course_name, credits)

-> Enrollment (enroll_id, Student_id, course_id, semester)

-> Grade {grade_id, enroll_id, marks)

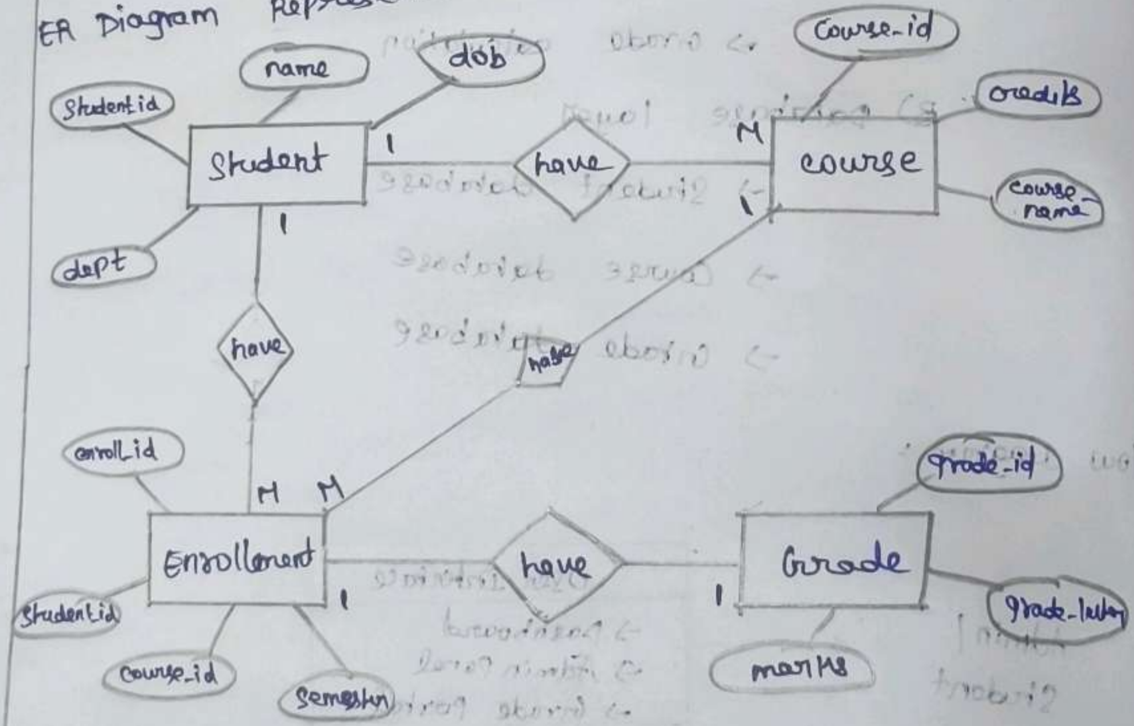
Relationship :

* Student -> Enrollment (1:N)

* course -> Enrollment (1:N)

* Enrollment -> Grade (1:1)

ER Diagram Representation:



Result:

Architectural & ER diagrams are designed successfully.

EXPNO : 8 Test Plans And Test Cases

AIM:

To create test Plans & test case for system validation

Design procedure:

1) Identify features
2) Define user action
3) Create happy path cases
4) create error cases
5) Assign test case IDs

User Stories:

-> Admin can add / update student
-> Student can enroll in course
-> Admin can assign grades
-> Student can view / download report

Test cases:

Test Suite : Student management

TC01 : Add Student
-> Action : Enter valid details
-> Expected : Student added successfully

TC02 : Add Student
-> Action : Submit empty fields
-> Expected : Error message displayed.

Test Suite : Course Enrollment.

TC003 : Enroll courses

-> Action : select course & enroll

-> Expected : Enrollment successfully

TC04 : Enroll course

-> Action : course full

-> Expected : "course is full" message

Test Suite : Grading

TC05 : Assign Grade

-> Action : Enter valid marks

-> Expected : Grade calculated correctly

TC06 : Assign Grade

Action : Enter invalid marks

Expected : Error message

Test Suite : Report card

TC07 : Download Report

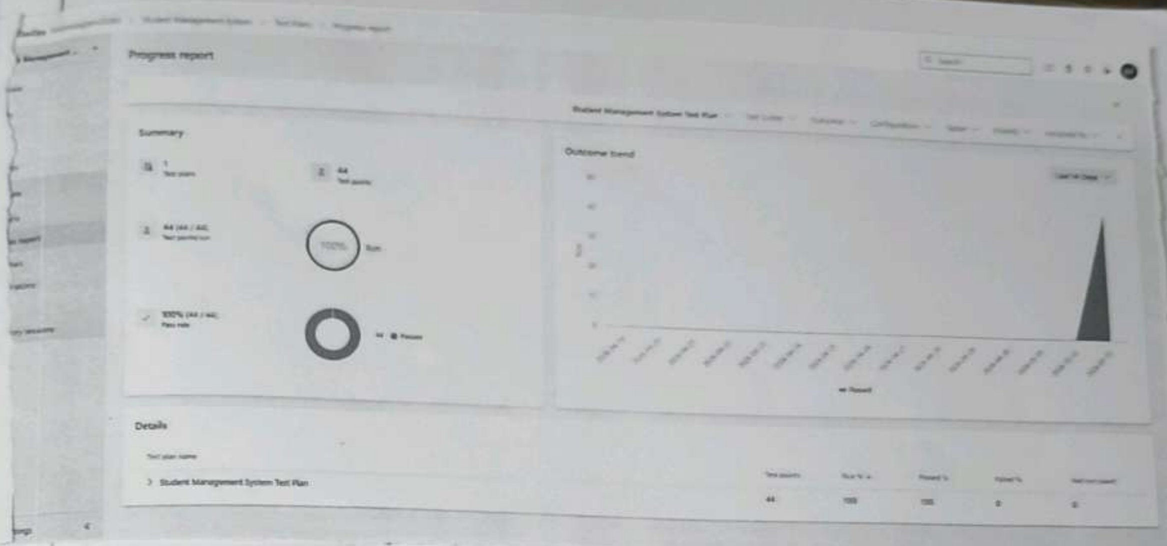
Action : click download

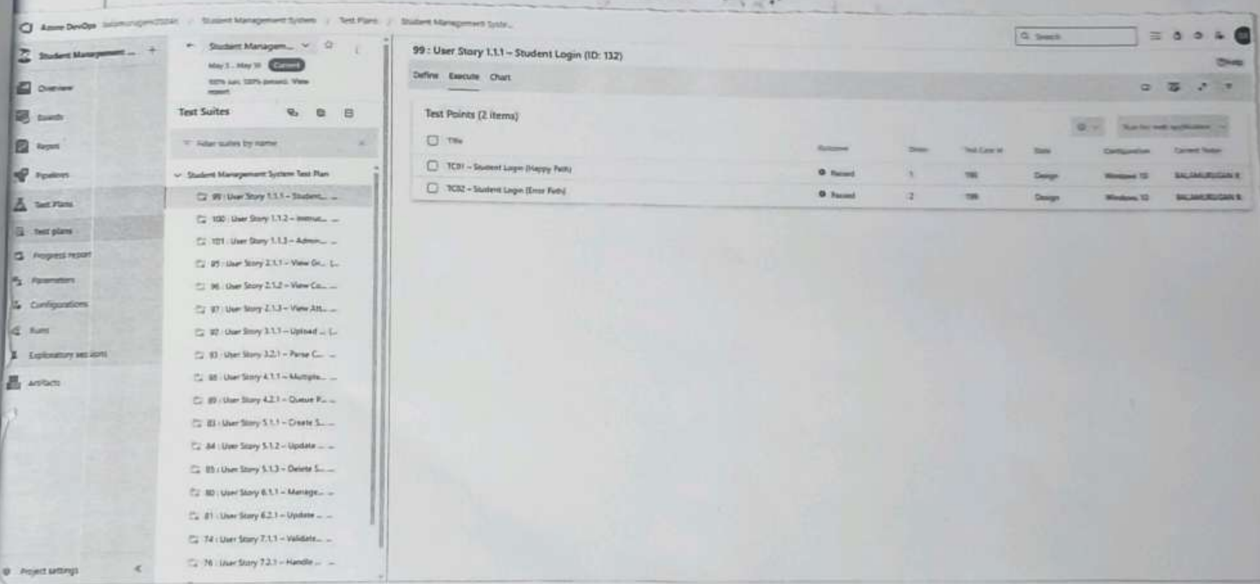
Expected : PDF generated.

TC08 : Download Report

Action : Grades not published

Expected : "Grades not available" message.





Result:

Test plans & test cases are created successfully with both happy & error scenarios

EXNO: 9 LOAD TESTING AND PIPELINES

AIM:

To perform load testing on the Student data Management System using Azure Load Testing & implement an Azure DevOPS pipeline to automate build & execution

Load Testing:

Description:

Azure Load Testing Simulates multiple users accessing the Student Data management System (Student records, attendance, grades, reports). It help analyze performance, scalability & system behavior under load

Steps to create Azure Load Testing Resource:

1) Login to Azure Portal (https://portal.azure.com)
2) Go to Create a Resource -> Azure Load Testing -> create
3) Enter details : Subscription, Resource Group, Name, Location
4) Click Review + create -> create
5) After deployment, open the resource

Steps to Run Load test:

1) Go to Load Testing resource
2) click Add HTTP Request -> create
3) Enter :
   -> Test name : Student -Test
   -> Description : Student System test
   -> Test URL : (Application endpoint)
4) click Create to run the test

Observation :

1) system handled multiple student requests effectively

2) Response time remained stable under load

3) Throughput increased with no. of users

4) No major errors observed

5) System showed good scalability.

PIPELINES :

Description :

Azure DevOps pipeline automates build & testing of the Student Data management system, ensuring smooth execution after every code output.

Steps :

1) connect GitHub :

-> create project in Azure DevOps

-> create pipeline & link GitHub repository.

2) Create YAML File :

```
trigger :
    - main
pool :
    vmImage : ubuntu - latest

Steps :
    - checkout : Self
    - task : UsePythonVersion@0
      inputs :
        versionSpec : '3.x'
    - script : pip install -r requirement.txt
    -> script : python -c "print('Pipeline Running')"
```

3) Run pipelines :

→ commit code → pipeline triggers automatically

→ Check logs in Azure DevOps

Observation (Pipelines)

1) pipelines executed successfully on commit
2) Dependencies installed correctly
3) All steps completed without errors
4) Logs confirmed proper execution
5) continuous integration achieved.

Result:

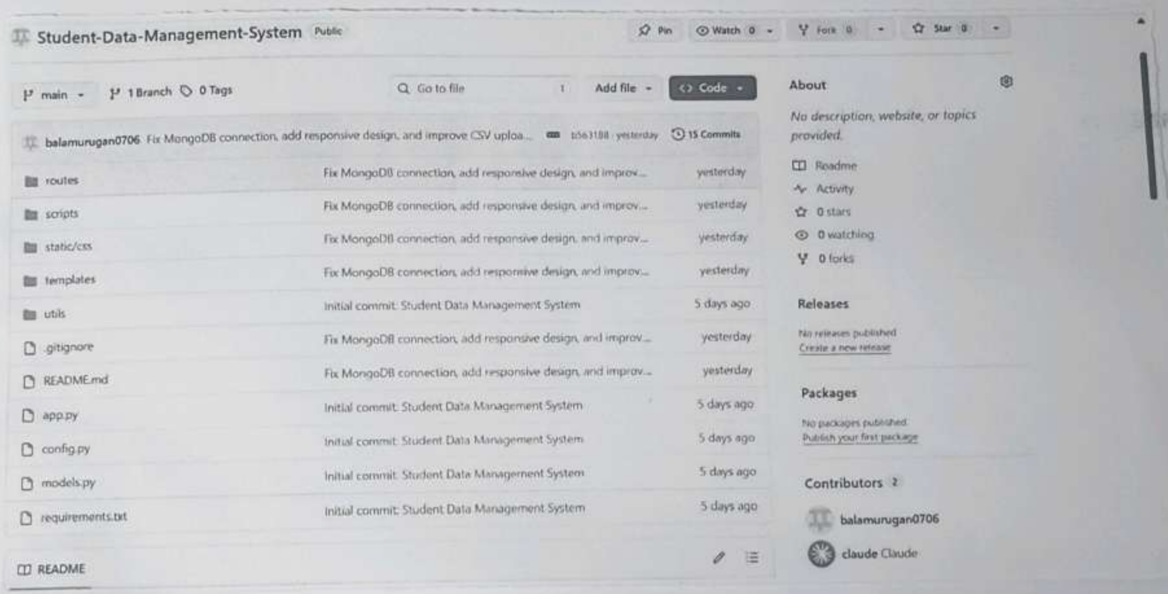
Successfully created Azure load testing resource and created load testing on the Student Data management system & demonstrated pipelines in Azure DevOps

EXPNO:10 GITHUB PROJECT STRUCTURE & NAMING CONVENTIONS

AIM:

To Provide a clear & organized view of the Student Data Management System Project Structure and file naming conventions.

Github project Structures:



Result:

The Project repository clearly Presents a well-Structured Folder organization & consistent naming conventions, making it easy to understand, access, and manage the Student Data Management System Code base